

BARATH RAJ S

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PROFESSIONAL SUMMARY

Final-year B.Tech (CCE) student with a CGPA of 8.10, seeking an entry-level AI/ML Engineer role. Hands-on experience building an AI-based video surveillance system using Python and OpenCV, and a flight delay prediction model using Scikit-learn. Proficient in Python, Machine Learning, Deep Learning fundamentals, and SQL. Completed NPTEL Python for Data Science and IBM Data Science Tools certifications. Currently developing skills in LLMs, RAG pipelines, and cloud deployment.

TECHNICAL SKILLS

Programming Languages: Python, Java
ML / AI Libraries: Scikit-learn, NumPy, Pandas, Matplotlib, OpenCV
Web Technologies: HTML, CSS, JavaScript
Database: SQL (MySQL)
Tools & Platforms: Git, GitHub, Jupyter Notebook, MS Office Suite
Concepts: Machine Learning, Computer Vision, Data Preprocessing, Feature Engineering, Model Evaluation

EDUCATION

B.Tech – Computer and Communication Engineering	2022 – 2026
<i>Sri Manakula Vinayagar Engineering College, Puducherry</i>	CGPA: 8.05
Higher Secondary Certificate (HSC)	2020 – 2022
<i>Jeevanandham Govt. Boys Higher Secondary School, Puducherry</i>	76.7%
Secondary School Leaving Certificate (SSLC)	2019 – 2020
<i>Thamizh Tendral Thriu.Vi.Ka Govt. Boys High School, Puducherry</i>	54.6%

PROJECTS

Smart AI-Based Video Surveillance System

- Built a real-time AI surveillance system using Python and OpenCV, achieving object detection with YOLOv8 on live video feed.
- Implemented automatic frame capture of suspicious activity and integrated Telegram Bot API for instant alerts, reducing response time significantly.
- Applied object detection and motion analysis techniques; achieved reliable real-time performance on standard hardware.
- Technologies:** Python, OpenCV, Machine Learning, Artificial Intelligence, Telegram API

PDF RAG Chatbot Python Mini Project

- Developed a Retrieval-Augmented Generation (RAG) chatbot that enables users to upload PDF documents and ask context-aware questions using natural language.
- Built a semantic search pipeline using FAISS vector database and HuggingFace sentence-transformers (all-MiniLM-L6-v2) for document chunking and embedding.
- Integrated Groq LLM (LLaMA 3.1-8B) for fast, context-grounded answer generation with source-page attribution displayed in the UI.
- Deployed an interactive Streamlit web interface with end-to-end PDF ingestion, vector indexing, and real-time Q&A; functionality.
- Technologies:** Python, Lang Chain, FAISS, HuggingFace Embeddings, Groq LLM, Streamlit, PyPDF

Flight Delay Prediction – ML Model

- Built a supervised ML model to predict flight delays using historical airline data (100K+ records), achieving 82%+ classification accuracy.
- Performed end-to-end pipeline: data cleaning, missing-value imputation, label encoding, feature engineering, and cross-validation.
- Evaluated and compared multiple algorithms (Logistic Regression, Random Forest, XGBoost); selected best model based on F1-score.
- **Technologies:** Python, Pandas, Scikit-learn, Matplotlib, Machine Learning

INTERNSHIP EXPERIENCE

Java Developer Intern – Sysslan IT Solutions

May 2026

- Collaborated with the development team on Java-based application modules, contributing to feature implementation and bug resolution.
- Applied Object-Oriented Programming (OOP) principles — encapsulation, inheritance, and polymorphism — in a professional codebase.
- Gained exposure to agile team workflows, code reviews, and version control practices.
- **Technologies:** Java, OOP, Problem Solving

CERTIFICATIONS

- Python for Data Science – NPTEL (IIT Madras)
- Data Science Tools Analyst – IBM SkillBuild